

Problem Set 6

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Problem 1

As noted in equation 15.1.10, the potential is

$$V(r) = \frac{1}{2} - \frac{GM}{r} + \frac{\ell^2}{2r^2} - \frac{GM\ell^2}{r^3}.$$

Following the notes, we choose units such that $GM = 1$, yielding potential

$$V(r) = \frac{1}{2} - \frac{1}{r} + \frac{\ell^2}{2r^2} - \frac{\ell^2}{r^3}.$$

Then

$$V'(r) = \frac{1}{r^2} - \frac{\ell^2}{r^3} + \frac{3\ell^2}{r^4}.$$

We can now directly solve the differential equations.

```
1 (* basic parameters *)
2 ell = 10;
3 rPlus = 1/2 (ell^2 + ell Sqrt[ell^2 - 12]);
4 r0 = 8 rPlus;
5
6 (* -V'(r) *)
7 rhs[r_] := -1/r^2 + ell^2/r^3 - (3 ell^2)/r^4;
8
9 (* have to wait till r'=0 the second time; the first time is its closest approach *)
10 turnCount = 0;
11
12 (*solve the differential equation for like a few orbits *)
13 sol = NDSolve[{
14   r'[lambda] == rhs[r[lambda]],
15   phi'[lambda] == ell/r[lambda]^2,
16   r[0] == r0,
17   r'[0] == 0,
18   phi[0] == 0
19 },
20 {r, phi},
21 {lambda, 0, 300000},
22 Events -> {
23   WhenEvent[r'[lambda] == 0 && lambda > 10,
24     {turnCount++, If[turnCount == 2, lambdaEnd = lambda]}
25   ]
26 }
27 ];
28
29 rFun = r /. sol[[1]];
30 phiFun = phi /. sol[[1]];
31
32 phiFinal = phiFun[lambdaEnd];
33 precession = phiFinal - 2 Pi;
34 Print["Precession Angle: ", precession];
```

We obtain a precession angle of 0.206374.

```
1
2 (* r(lambda) *)
3 Plot[rFun[1], {1, 0, 5.5 * lambdaEnd},
4 AxesLabel -> {"\[Lambda]", "r"},
5 PlotStyle -> Blue,
6 GridLines -> Automatic]
7
8 (* phi(lambda) *)
9 Plot[phiFun[1], {1, 0, 5.5 * lambdaEnd},
10 AxesLabel -> {"\[Lambda]", "\[Phi]"},
11 PlotStyle -> Darker[Green],
12 GridLines -> Automatic]
13
14 (* parametric orbit *)
15 ParametricPlot[{rFun[1] Cos[phiFun[1]], rFun[1] Sin[phiFun[1]]},
16 {1, 0, 5.5 * lambdaEnd},
17 AxesLabel -> {"x", "y"},
18 PlotStyle -> Black,
19 PlotRange -> All,
20 AspectRatio -> 1,
21 GridLines -> Automatic]
```

The figures are on another page because LaTeX formatting sucks.

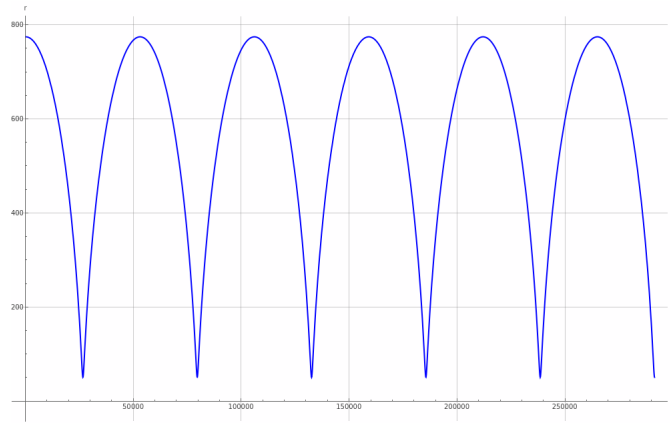


Figure 1: $r(\lambda)$ plotted against λ .

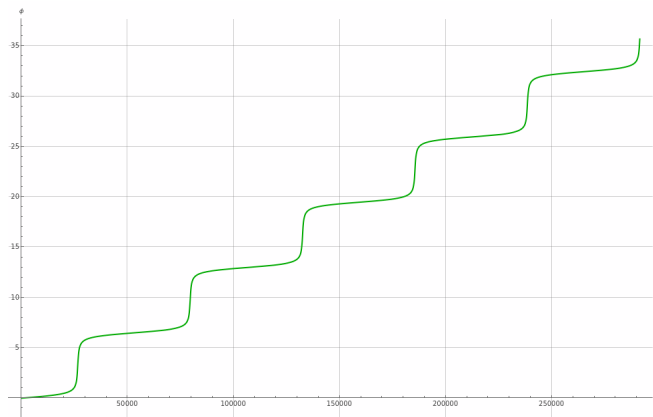


Figure 2: $\phi(\lambda)$ plotted against λ .

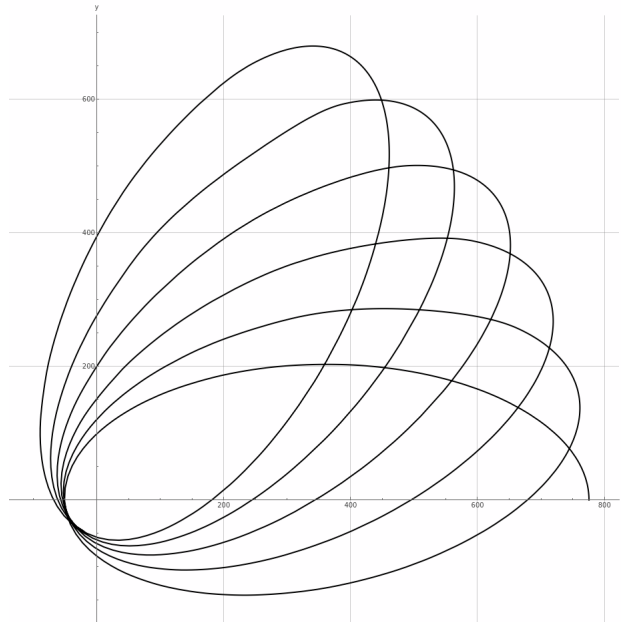


Figure 3: Parametric plot of the orbit.